PRECALCULUS · UNIT 7: VECTORS

Flight Path Navigator

Use vectors to fly a small plane through wind, find ground speed, and plan a delivery route.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

You are the flight planner for Coastal Air Delivery. Small planes fly between islands, and the wind pushes them off course. Pilots need to know which way to point the plane and how long each leg will take.

You will use vectors to plan safe, efficient flights.



Plane

Airspeed 150 mph

Wind

30 mph from the west

(blowing east)

Learning Goals

- I can add and scale vectors and find magnitude and direction.
- I can find the heading that corrects for wind.
- I can use the dot product to find angles and work.

Project Timeline

When	What you will do
Day 1	Route Vectors
Day 2	Wind Correction
Day 3	Cargo & Work
Final	Flight Plan + reflection

Part 1: Route Vectors

Day 1 · Use the Flight Data Cards.

Flight Data Cards (x = east, y = north, miles)

Mainland Airport	Gull Island	Pine Island
(0, 0)	(120, 90)	(200, −50)

Wind A	Wind B	Cargo Ramp
⟨30, 0⟩ mph	25 mph toward bearing 135° (SE)	Pull 80 lb at 25° for 40 ft

1. Find the vectors and distances for Mainland → Gull → Pine → Mainland.

2. Show that the three vectors add to the zero vector. Why?

3. Find the bearing (clockwise from north) from Mainland to Gull Island.

4. Find the angle at Gull Island between the incoming and outgoing legs using the dot product.

Part 2: Wind Correction

Day 2 · Airspeed 150 mph; wind ⟨30, 0⟩. Fly from Mainland to Gull Island.

5. If the pilot aims straight at Gull Island, what is the ground velocity and where does the plane actually go?

6. Find the heading so the ground track points at Gull Island. (Let the air velocity be $150\langle\cos\theta, \sin\theta\rangle$.)

7. Find the ground speed with that heading and the flight time.

8. Explain why the corrected heading points into the wind.

Part 3: Cargo & Work

Day 3 · Use the dot product: W = F · d.

9. A worker pulls a cart with 80 lb of force at 25° above the ramp for 40 ft. Find the work.

10. Are ⟨3, 4⟩ and ⟨−8, 6⟩ perpendicular? Use the dot product.

11. Write your flight plan for the full route with headings, ground speeds, and times.

Final Product: Flight Plan

Your final product must include:

- Route vectors and closure
- Bearings and angles via dot product
- Wind-corrected heading and time
- Work calculation
- Flight plan

Reflection

R1. How does the dot product reveal the angle between two vectors?

R2. What is the difference between heading and course?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Vectors, resultants, headings, and dot products are computed correctly and produce a workable flight plan. Goes beyond the minimum with a creative, realistic choice.	Vectors, resultants, headings, and dot products are computed correctly and produce a workable flight plan.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Flight plan includes diagrams, components, headings, speeds, and times. Polished, organized, and visually engaging.	Flight plan includes diagrams, components, headings, speeds, and times. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.